$C_{\phi D}^{(22)}\ {}^{(22)} \to \frac{1}{16\,\pi^2}$

$$\Big(-\tfrac{1}{54}\sum_p g_1^4\, LF_{3,0}[m_{\tilde d}^p] + \tfrac{5}{144}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde d}^p] - \tfrac{2}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde d}^p] - \tfrac{1}{18}\sum_p g_1^4\, LF_{3,0}[m_{\tilde e}^p] +$$

$$\tfrac{5}{48}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde e}^p] - \tfrac{2}{45}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde e}^p] + \Big(\tfrac{1}{4} g_2^2\, \overline{y_e}^{pr} y_e^{pr} - \tfrac{1}{144}\sum_p \big(4 g_1^4 + 9 g_2^4\big)\Big)$$

$$LF_{3,0}[m_{\tilde l}^p] + \tfrac{1}{96}\Big(-24 g_2^2\, \overline{y_e}^{pr} y_e^{pr} + \sum_p \big(5 g_1^4 + 6 g_2^4\big)\Big) LF_{4,-1}[m_{\tilde l}^p] -$$

$$\tfrac{1}{45}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde l}^p] + \Big(\tfrac{3}{4} g_2^2\, \overline{y_d}^{pr} y_d^{pr} - \tfrac{1}{432}\sum_p \big(4 g_1^4 + 81 g_2^4\big)\Big) LF_{3,0}[m_{\tilde q}^p] +$$

$$\Big(-\tfrac{3}{4} g_2^2\, \overline{y_d}^{pr} y_d^{pr} + \tfrac{1}{288}\sum_p \big(5 g_1^4 + 54 g_2^4\big)\Big) LF_{4,-1}[m_{\tilde q}^p] - \tfrac{1}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde q}^p] -$$

$$\tfrac{2}{27}\sum_p g_1^4\, LF_{3,0}[m_{\tilde u}^p] + \tfrac{5}{36}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde u}^p] - \tfrac{8}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde u}^p] - \tfrac{1}{36} g_1^4\, LF_{3,0}[\tilde\mu] -$$

$$\tfrac{1}{24} g_1^4\, LF_{4,-1}[\tilde\mu] + \tfrac{2}{45} g_1^4\, LF_{5,-2}[\tilde\mu] - \tfrac{1}{32} g_1^4\, LF_{2,2,-1}[m_1, \tilde\mu] - \tfrac{1}{16} g_1^4 m_1^2\, LF_{2,2,0}[m_1, \tilde\mu] -$$

$$\tfrac{17}{32} g_2^4\, LF_{2,2,-1}[m_2, \tilde\mu] - \tfrac{1}{16} g_2^4 m_2^2\, LF_{2,2,0}[m_2, \tilde\mu] + g_2^4\, LF_{3,2,-2}[m_2, \tilde\mu] -$$

$$\tfrac{3}{4} g_2^4 m_2^2\, LF_{3,2,-1}[m_2, \tilde\mu] - \tfrac{1}{2} g_2^4\, LF_{3,3,-3}[m_2, \tilde\mu] + \tfrac{1}{2} g_2^4 m_2^2\, LF_{3,3,-2}[m_2, \tilde\mu] -$$

$$\tfrac{1}{2} g_2^4\, LF_{4,2,-3}[m_2, \tilde\mu] + \tfrac{1}{2} g_2^4 m_2^2\, LF_{4,2,-2}[m_2, \tilde\mu] - \tfrac{1}{6} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{2,2,0}[m_{\tilde d}^r, m_{\tilde q}^p] +$$

$$\tfrac{1}{12} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{3,2,-1}[m_{\tilde d}^r, m_{\tilde q}^p] - \tfrac{1}{6} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{2,2,0}[m_{\tilde e}^r, m_{\tilde l}^p] +$$

$$\tfrac{1}{12} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{3,2,-1}[m_{\tilde e}^r, m_{\tilde l}^p] + \tfrac{1}{6} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{3,1,0}[m_{\tilde l}^p, m_{\tilde e}^r] +$$

$$\tfrac{1}{6} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{3,2,-1}[m_{\tilde l}^p, m_{\tilde e}^r] + \tfrac{1}{8}\, \overline{a_e}^{pr} a_e^{pr} \big(-3 g_1^2 + g_2^2\big)\, LF_{4,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r] +$$

$$\tfrac{1}{6} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{5,1,-2}[m_{\tilde l}^p, m_{\tilde e}^r] - \tfrac{1}{4}\, \overline{y_e}^{pr}\, \overline{y_e}^{st} y_e^{pt} y_e^{sr}\, LF_{2,1,0}[m_{\tilde l}^p, m_{\tilde l}^s] +$$

$$\tfrac{1}{4}\, \overline{y_e}^{pr}\, \overline{y_e}^{st} y_e^{pt} y_e^{sr}\, LF_{3,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s] + \tfrac{1}{6} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{3,1,0}[m_{\tilde q}^p, m_{\tilde d}^r] +$$

$$\tfrac{1}{6} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{3,2,-1}[m_{\tilde q}^p, m_{\tilde d}^r] + \tfrac{1}{8}\, \overline{a_d}^{pr} a_d^{pr} \big(-5 g_1^2 + 3 g_2^2\big)\, LF_{4,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r] +$$

$$\tfrac{1}{2} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{5,1,-2}[m_{\tilde q}^p, m_{\tilde d}^r] - \tfrac{3}{4}\, \overline{y_d}^{pr}\, \overline{y_d}^{st} y_d^{pt} y_d^{sr}\, LF_{2,1,0}[m_{\tilde q}^p, m_{\tilde q}^s] +$$

$$\tfrac{3}{4}\, \overline{y_d}^{pr}\, \overline{y_d}^{st} y_d^{pt} y_d^{sr}\, LF_{3,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s] + \tfrac{1}{12} g_1^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{2,2,0}[m_{\tilde q}^p, m_{\tilde u}^r] -$$

$$\tfrac{3}{4} g_2^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{3,1,0}[m_{\tilde q}^p, m_{\tilde u}^r] - \tfrac{1}{24} \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr} \big(g_1^2 - 9 g_2^2\big)\, LF_{3,2,-1}[m_{\tilde q}^p, m_{\tilde u}^r] +$$

$$\tfrac{3}{4} g_2^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{4,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r] - \tfrac{1}{12} g_1^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{3,1,0}[m_{\tilde u}^r, m_{\tilde q}^p] -$$

$$\tfrac{1}{12} g_1^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{3,2,-1}[m_{\tilde u}^r, m_{\tilde q}^p] - \tfrac{1}{4} g_1^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{4,1,-1}[m_{\tilde u}^r, m_{\tilde q}^p] +$$

$$\tfrac{1}{2} g_1^2 \tilde\mu^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{5,1,-2}[m_{\tilde u}^r, m_{\tilde q}^p] - \tfrac{1}{16} g_1^4\, LF_{3,2,-2}[\tilde\mu, m_1] +$$

$$\tfrac{1}{16} g_1^4 m_1^2\, LF_{3,2,-1}[\tilde\mu, m_1] + \tfrac{5}{24} g_1^4\, LF_{4,1,-2}[\tilde\mu, m_1] + \tfrac{1}{16} g_1^4\, LF_{4,2,-3}[\tilde\mu, m_1] -$$

$$\tfrac{1}{16} g_1^4 m_1^2\, LF_{4,2,-2}[\tilde\mu, m_1] - \tfrac{1}{6} g_1^4\, LF_{5,1,-3}[\tilde\mu, m_1] + \tfrac{3}{16} g_2^4\, LF_{3,2,-2}[\tilde\mu, m_2] +$$

$$\tfrac{1}{16} g_2^4 m_2^2\, LF_{3,2,-1}[\tilde\mu, m_2] + \tfrac{5}{8} g_1^2 g_2^2\, LF_{4,1,-2}[\tilde\mu, m_2] + \tfrac{1}{16} g_2^4\, LF_{4,2,-3}[\tilde\mu, m_2] -$$

$$\tfrac{1}{16} g_2^4 m_2^2\, LF_{4,2,-2}[\tilde\mu, m_2] - \tfrac{1}{2} g_1^2 g_2^2\, LF_{5,1,-3}[\tilde\mu, m_2] + g_1^2 g_2^2\, LF_{2,2,1,-2}[m_2, \tilde\mu, m_1] +$$

$$\tfrac{3}{4} m_1 m_2 g_1^2 g_2^2\, LF_{2,2,1,-1}[m_2, \tilde\mu, m_1] - \tfrac{1}{2} g_1^2 g_2^2\, LF_{3,2,1,-3}[m_2, \tilde\mu, m_1] -$$

$$\tfrac{1}{2} m_1 m_2 g_1^2 g_2^2\, LF_{3,2,1,-2}[m_2, \tilde\mu, m_1] - \tfrac{1}{4}\, \overline{y_e}^{st} y_e^{pt}\, \overline{a_e}^{pr} a_e^{sr}\, LF_{2,2,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r] -$$

$$\tfrac{3}{4}\, \overline{y_d}^{st} y_d^{pt}\, \overline{a_d}^{pr} a_d^{sr}\, LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r] + \tfrac{3}{4} \tilde\mu^2\, \overline{y_d}^{pr} y_d^{sr}\, \overline{y_u}^{st} y_u^{pt}\, LF_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] -$$

$$\tfrac{3}{4} \tilde\mu^2\, \overline{y_d}^{pr} y_d^{sr}\, \overline{y_u}^{st} y_u^{pt}\, LF_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] -$$

$$\tfrac{3}{8} \tilde\mu^2\, \overline{y_d}^{pr} y_d^{sr}\, \overline{y_u}^{st} y_u^{pt}\, LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s] + \tfrac{3}{4} \tilde\mu^2\, \overline{y_d}^{pr} y_d^{sr}\, \overline{y_u}^{st} y_u^{pt}$$

$$LF_{2,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^t] - \tfrac{3}{4} \tilde\mu^2\, \overline{y_d}^{pr} y_d^{sr}\, \overline{y_u}^{st} y_u^{pt}\, LF_{3,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^t] -$$

$$\tfrac{3}{8} \tilde\mu^2\, \overline{y_d}^{pr} y_d^{sr}\, \overline{y_u}^{st} y_u^{pt}\, LF_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde q}^p] - \tfrac{7}{16} g_1^2 g_2^2\, LF_{2,1,1,-1}[\tilde\mu, m_1, m_2] -$$

$$\tfrac{1}{8} m_1 m_2 g_1^2 g_2^2\, LF_{2,1,1,0}[\tilde\mu, m_1, m_2] + \tfrac{3}{8} g_1^2 g_2^2\, LF_{3,1,1,-2}[\tilde\mu, m_1, m_2] +$$

$$\tfrac{1}{8} m_1 m_2 g_1^2 g_2^2\, LF_{3,1,1,-1}[\tilde\mu, m_1, m_2] - \tfrac{1}{8} g_1^2 g_2^2\, LF_{4,1,1,-3}[\tilde\mu, m_1, m_2] -$$

$$\tfrac{1}{8} m_1 m_2 g_1^2 g_2^2\, LF_{4,1,1,-2}[\tilde\mu, m_1, m_2] - \tfrac{1}{2} g_1^2 g_2^2\, LF_{3,2,1,-3}[\tilde\mu, m_2, m_1] -$$

$$\tfrac{1}{2} m_1 m_2 g_1^2 g_2^2\, LF_{3,2,1,-2}[\tilde\mu, m_2, m_1] - \tfrac{1}{8}\, \overline{a_e}^{pr}\, \overline{a_e}^{st} a_e^{pt} a_e^{sr}\, LF_{2,2,1,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r, m_{\tilde e}^t] -$$

$$\tfrac{3}{8}\, \overline{a_d}^{pr}\, \overline{a_d}^{st} a_d^{pt} a_d^{sr}\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde d}^t] +$$

$$\tfrac{3}{4} \tilde\mu^2\, \overline{y_u}^{st} y_u^{pt}\, \overline{a_d}^{pr} a_d^{sr}\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde u}^t] -$$

$$\tfrac{3}{4} \tilde\mu^4\, \overline{y_u}^{pr}\, \overline{y_u}^{st} y_u^{pt} y_u^{sr}\, LF_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] +$$

$$\tfrac{3}{4} \tilde\mu^4\, \overline{y_u}^{pr}\, \overline{y_u}^{st} y_u^{pt} y_u^{sr}\, LF_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] +$$

$$\tfrac{3}{8} \tilde\mu^4\, \overline{y_u}^{pr}\, \overline{y_u}^{st} y_u^{pt} y_u^{sr}\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] +$$

$$\tfrac{3}{8} \tilde\mu^4\, \overline{y_u}^{pr}\, \overline{y_u}^{st} y_u^{pt} y_u^{sr}\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s, m_{\tilde u}^r]\Big)$$